Taller #3

1)

a) 7 obj. en 3 espa.

$V_r(\underset{7}{n},\underset{3}{x}) = n^x = 7^3 = 343$ Rta = 343

b) 9 obj en 8 esp.

$V_r(9,8) = 9^8 = 43046721$ Rta = 43046721

c) 8 obj en 4 esp. dif.

$P(8,\cancel{4}) = \frac{8!}{(8-4)!} = \frac{8!}{4!} = \frac{8\cdot7\cdot6\cdot5\cdot\cancel{4}}{\cancel{4}} = 1680$ = Rta 1680.

d) 6 obj. en 2 esp. dif.

$P(6,2) = \frac{6!}{(6-2)!} = \frac{6!}{4!} = \frac{6\cdot5\cdot\cancel{4}}{\cancel{4}} = 30$ Rta = 30

2)

a) bookkeeper

~~b~~	b	1
~~o~~	o	2
~~k~~	k	2
~~k~~	e	3
	p	1
	r	1

$$p = \frac{n!}{a!b!c!} = \frac{10!}{1!2!2!3!1!1!}$$

~~10 · 9 · 8 · 7 · 6 · 5 · 4 · 3 · 2~~

$10! = 3628800$

$2! = 2$

$3! = 6$

$$\frac{3628800}{1 \cdot 2 \cdot 2 \cdot 6 \cdot 1} = \frac{3628800}{24} = 151200 = \text{Rta}$$

b)

12 discos

↓

5 n

3 r

2 b

2 v

$$\frac{12!}{5!3!2!2!} = \frac{479001600}{120 \cdot 6 \cdot 2 \cdot 2} = \frac{479001600}{2880} =$$

$$= 166320 = \text{Rta.}$$

c)

6 preg. de 10

$$C(n,r) = \frac{n!}{(n-r)!r!} = \frac{10!}{(10-6)!6!} = \frac{10!}{4!6!} = \frac{3628800}{24 \cdot 720}$$

$$\frac{3628800}{17280} = 210 \quad \text{Rta} = 210$$

d)

3 8 puntos

$C(8,3) = 56$. Rta=56

e)

4 faldos

↓ $4 \cdot 6 = 24$ Rta= 24

6 blusas.

3)

a)

~~3 Cent.~~

~~10~~

Central

Def.

Atq.

Extr.

Halfb.

Mosis.

Full b.

b) 3 h.   12 h.

2 m.   8 m.

$c(12,3) = 220$

$220 \cdot 28 = 6160$ Rta=6160

$c(8,2) = 28$